
Ku-band 8-channel Transmit Multifunctional Integrated Chip

(TRHJ-2041)

User Manual

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Product Features

Operating frequency range: 13.7GHz to 14.5GHz
 8 independent controllable transmitting channels
 Output 1dB power of compression point (P1dB): +13dBm
 Gain: +19dB
 Power added efficiency: 18%@P1dB
 360 °phase adjustment: 6-bit, 5.625 °;
 31dB amplitude adjustment: 6-bit, 0.5dB
 Integrated temperature sensing and temperature compensation
 Power detection range: 4dBm to 14dBm
 Programmed bias mode
 SPI bus: 4 wires
 Power consumption of chip (static): 0.22W
 Single-channel power consumption of chip (P1dB): 0.12W
 Packaging form: WLCSP

Applications

Satellite communications

Summarization

TRHJ-2041 is an 8-channel transmit beamforming phased array silica-based (CMOS) chip operating from 13.7GHz to 14.5GHz. The input signals are input from RF_IO of the common port, pass through eight transmitting channels via 8 power dividers and are output at RF_IO of the common port. The chip provides 6-bit phase (adjustment range: 360 °, step: 5.625 °) and 6-bit amplitude adjustment (adjustment range: 31dB, step: 0.5dB).

A simple four-line interface (SPI) is adopted for the TRHJ-2041 chip to control the on-chip register. In addition, it is acceptable for SPI to control at most 16 chips on the same serial line four address pins. A specialized transmitting load pin can also provide the synchronization of all core chips in the same array.

WLCSP capsulation is adopted for TRHJ-2041.

The operating environment temperature range is -55°C to +85°C

Functional block diagram

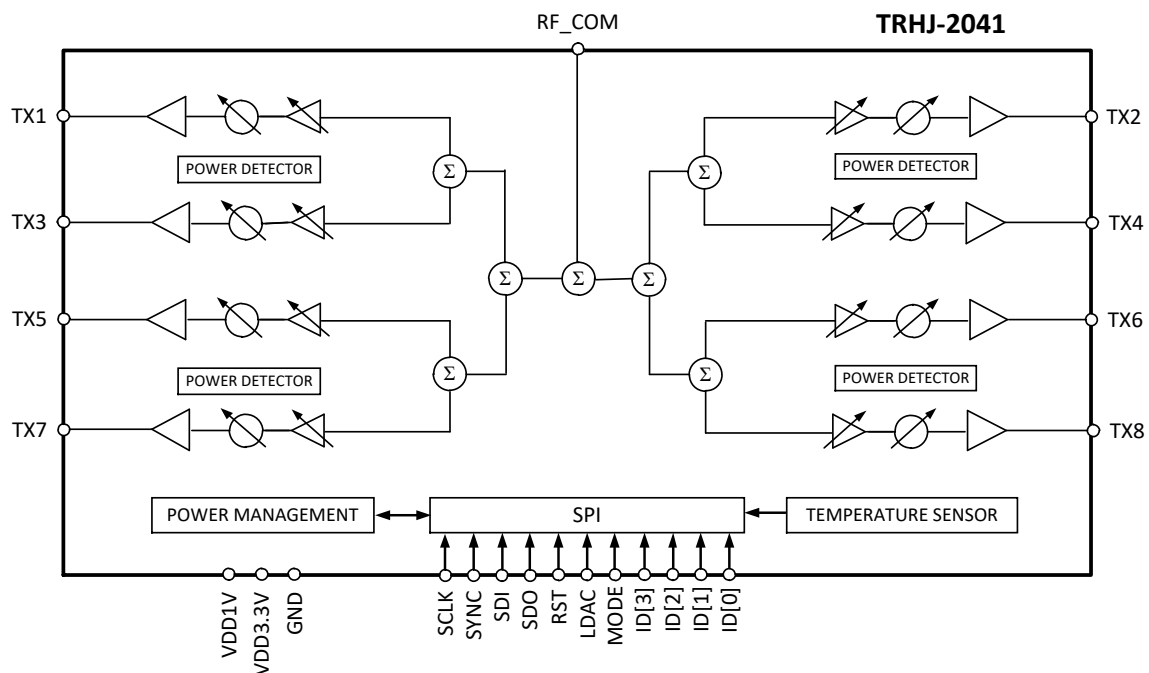


Figure 1 Functional Block Diagram of Chip

Technical Specifications

In the testing results below, all transmitting modes are set as the maximum gain status and standard bias condition. See the status setup of the bias register in digital chip control. VDD1=1V, VDD3.3=3.3V and TA=25 °C.

Table 1. Technical Specification of Chip

Parameter	Test conditions	Min.	Typical	Max.	Unit
Working conditions					
Frequency range		13.7	14.1	14.5	GHz
Working environment temperature		-55		+85	°C
Transmitting mode					
Maximum gain					
13.7 GHz			21		dB
14.1 GHz			20		dB
14.5 GHz			19.5		dB
Gain flatness			±0.75		dB
Gain varies with temperature	13.7GHz(-55°C to +105°C)		4.5		dB
Output 1 dB compression point (OP1dB)	Batch 70+ sample statistics				
13.7 GHz		13.4	13.8	14.2	dBm
14.1 GHz		13.35	13.9	14.3	dBm
14.5 GHz		13.3	13.7	14.1	dBm
Power added efficiency@1dB compression point	Batch 70+ sample statistics				
13.7 GHz		13.9	18.2	21	%
14.1 GHz		14.2	18.6	21.6	%
14.5 GHz		13.9	17.8	20.8	%
Amplitude adjustable range			31.5		dB
Amplitude modulation accuracy			0.5		dB
Root-mean-square (RMS) of amplitude modulation difference		0.3	0.4	0.5	dB
Adjustable phase range			360		Degree
Phase shift precision			5.625		Degree
Root-mean-square (RMS) of phase shifting difference		2.9	3	3.2	Degree
Output return loss	TX1, TX2, TX3,, TX8 pin	-10	-9	-7	dB
Input return loss	RF_COM pin	-20	-15	-13	dB
Isolation between channels					
Amplitude		0.35		1	dB
Phase		2.3		4.8	Degree
Operating current					
1V	Single-channel work		20		mA
3.3V	Single-channel work		2.1		mA

Temperature sensor					
Temperature readback range		-60		70	°C
Temperature compensation			8		Bits
Power detection range					
RF power		4		14	dBm

Absolute maximum rating value

Table 2. Absolute Maximum Rating Value

Parameter	Absolute maximum rating value	Unit
RF maximum input power	6	dBm
VDD1V	1.2	V
VDD3.3V	3.6	V
Operating case temperature range	-40°C - +85°C	°C
Storage temperature range	-55°C - +125°C	°C
Maximum junction temperature	135	°C
Pin temperature (welding, 60 seconds)	260	°C
Electrostatic discharge sensitivity (ESD)	500	V

Note: exceeding or equal to the above absolute maximum rating may cause permanent damage to the product. This is only the rated value and does not mean that the device can work normally under these conditions or under any other conditions beyond the specifications shown in the operation section of the Technical Specification. Long term operation beyond the maximum rated value will affect the reliability of the product.

	ESD (electrostatic discharge) sensitive device Charged devices and circuit boards may discharge without detection. Although this product has patented or proprietary protection circuit, the device may be damaged in case of high-energy ESD. Therefore, appropriate ESD precautions should be taken to avoid device performance degradation or function loss.
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Pin configuration and function description

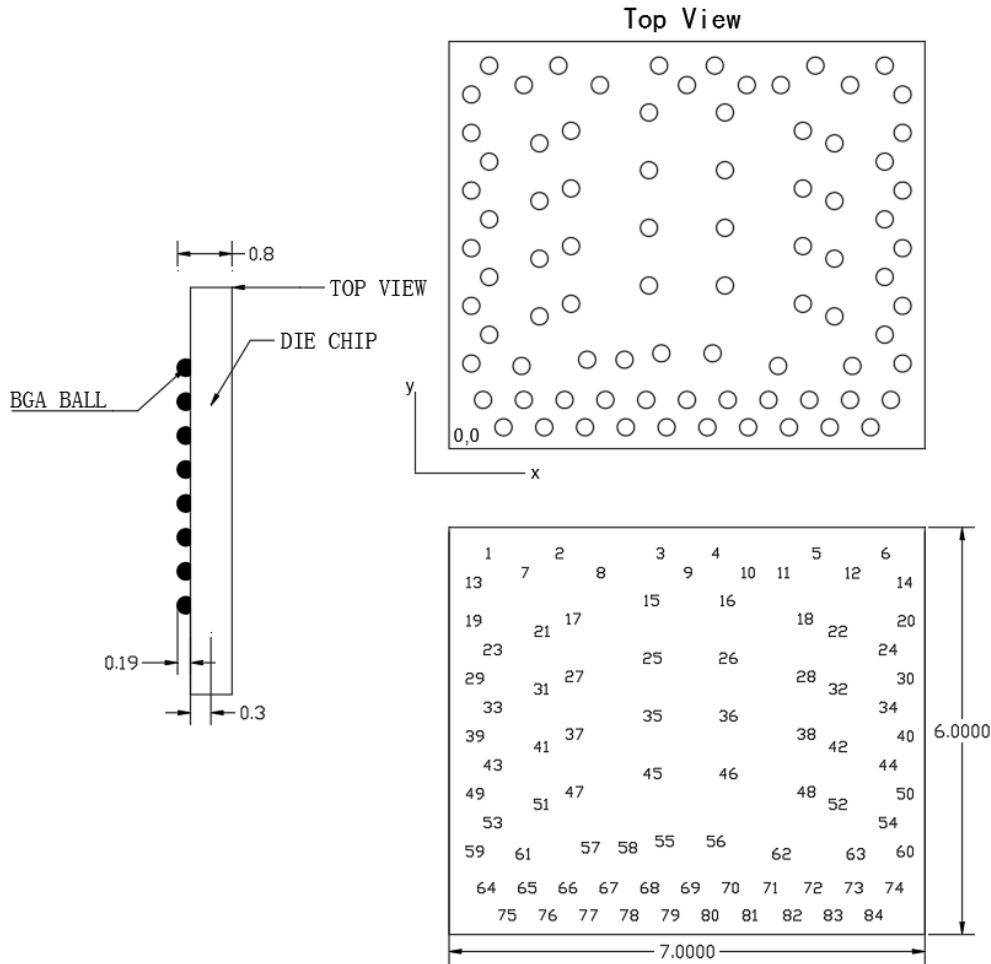


Figure 2. Pin Configuration Diagram

Table 3. Pin Function Description

Num berin g	RDL bump center		pad definition	Descriptio n	Num berin g	RDL bump center		pad definitio n	Description
	X coordinate	Y coordinat e				X coordinate	Y coordinate		
1	0.59	5.645	NULL	Suspended	43	0.5907	2.5297	SIG5	Channel 5
2	1.6096	5.645	NULL	Suspended	44	6.4093	2.5297	SIG4	Channel 4
3	3.0902	5.645	GND	grounding	45	2.941	2.4042	VDD10	RF 1V
4	3.9098	5.645	GND	grounding	46	4.0591	2.4042	VDD10	RF 1V
5	5.3904	5.645	NULL	Suspended	47	1.7945	2.1347	GND	grounding
6	6.41	5.645	NULL	Suspended	48	5.2055	2.1347	GND	grounding
7	1.0998	5.3585	NULL	Suspended	49	0.3274	2.1047	GND	grounding
8	2.2194	5.3585	NULL	Suspended	50	6.6727	2.1047	GND	grounding
9	3.5	5.3585	COM	RF joint	51	1.33	1.9498	VDD10	RF 1V
10	4.3806	5.3585	NULL	Suspended	52	5.6701	1.9498	VDD10	RF 1V
11	4.8806	5.3585	NULL	Suspended	53	0.5907	1.6797	SIG7	Channel 7
12	5.9002	5.3585	NULL	Suspended	54	6.4093	1.6797	SIG6	Channel 6

13	0.3274	5.2195	NULL	Suspended	55	3.1228	1.4047	SCLK	Control bit
14	6.6727	5.2195	NULL	Suspended	56	3.8772	1.4047	ID[0]	ID bit
15	2.941	4.9542	VDD10	RF 1V	57	2.0317	1.3108	ID[3]	ID bit
16	4.0591	4.9542	VDD10	RF 1V	58	2.5817	1.3108	ID[2]	ID bit
17	1.7945	4.6847	GND	grounding	59	0.3274	1.2547	GND	grounding
18	5.2055	4.6847	GND	grounding	60	6.6727	1.2547	GND	grounding
19	0.3274	4.6547	GND	grounding	61	1.0662	1.2188	MODE	Control bit
20	6.6727	4.6547	GND	grounding	62	4.8414	1.2188	NULL	Suspended
21	1.33	4.4998	VDD10	RF 1V	63	5.9338	1.2188	NULL	Suspended
22	5.6701	4.4998	VDD10	RF 1V	64	0.5	0.7188	DGND	grounding
23	0.5907	4.2297	SIG1	Channel 1	65	1.1	0.7188	DVDD	Digital 1V
24	6.4093	4.2297	SIG0	Channel 0	66	1.7	0.7188	3.3V	Digital 3.3V
25	2.941	4.1042	VDD10	RF 1V	67	2.3	0.7188	RST_IN	Control bit
26	4.0591	4.1042	VDD10	RF 1V	68	2.9	0.7188	SDO	Control bit
27	1.7945	3.8347	GND	grounding	69	3.5	0.7188	ID[1]	ID bit
28	5.2055	3.8347	GND	grounding	70	4.1	0.7188	GND	grounding
29	0.3274	3.8047	GND	grounding	71	4.7	0.7188	VDD33	Analog 3.3V
30	6.6727	3.8047	GND	grounding	72	5.3	0.7188	NULL	Suspended
31	1.33	3.6498	VDD10	RF 1V	73	5.9	0.7188	NULL	Suspended
32	5.6701	3.6498	VDD10	RF 1V	74	6.5	0.7188	NULL	Suspended
33	0.5907	3.3797	SIG3	Channel 3	75	0.8005	0.315	NULL	Suspended grounding Control bit
34	6.4093	3.3797	SIG2	Channel 2	76	1.4005	0.315	DGND	
35	2.941	3.2542	VDD10	RF 1V	77	2.0005	0.315	LDAC_N	
36	4.0591	3.2542	VDD10	RF 1V	78	2.6005	0.315	SYNC	Control bit
37	1.7945	2.9847	GND	grounding	79	3.2005	0.315	SDI	Control bit
38	5.2055	2.9847	GND	grounding	80	3.8005	0.315	VDD10	Analog 1V
39	0.3273	2.9547	GND	grounding	81	4.4005	0.315	NULL	Suspended
40	6.6727	2.9547	GND	grounding	82	5.0005	0.315	NULL	Suspended
41	1.33	2.7998	VDD10	RF 1V	83	5.6005	0.315	NULL	Suspended
42	5.6701	2.7998	VDD10	RF 1V	84	6.2005	0.315	NULL	Suspended

Typical Performance Parameters

Transmitting mode

In the testing results below, all transmitting modes are set as the maximum gain status and standard bias condition. See the status setup of the bias register in digital chip control. VDD1=1V, VDD3.3=3.3V and TA=25 ℃.

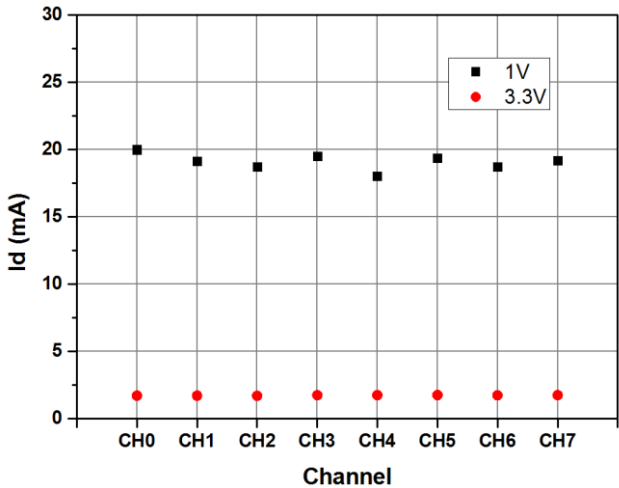


Figure 3. Operating Current under Single-Channel Status

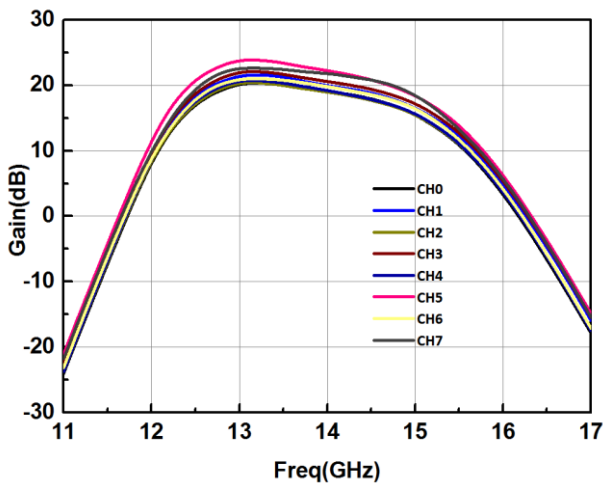


Figure 4. Relationship between Gain and Frequency with Different Channels

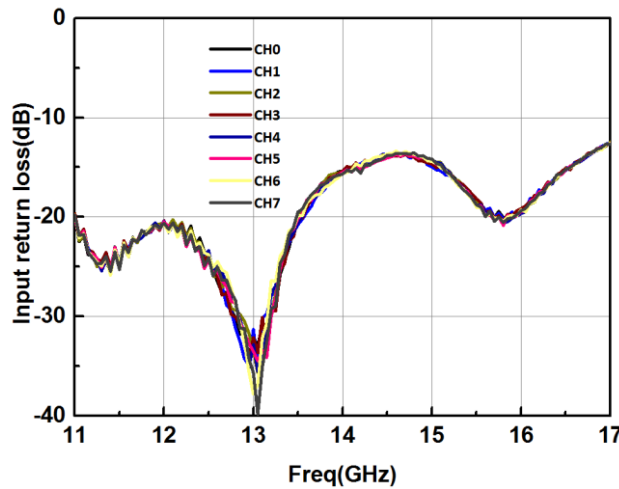


Figure 5. Relationship between Return Loss and Frequency with Different Channels

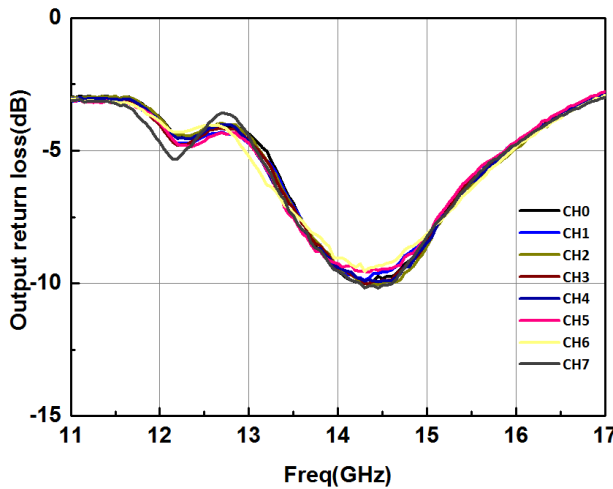


Figure 6. Relationship between Output 1dB Compression Point and Frequency with Different Channels

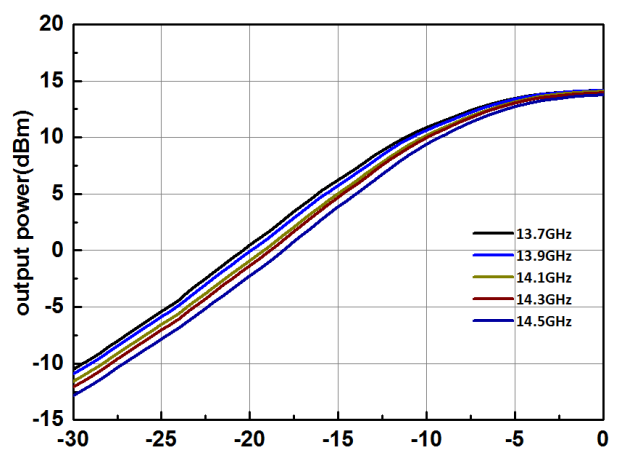


Figure 7. Relationship between Single-Channel Output Power and Input Power

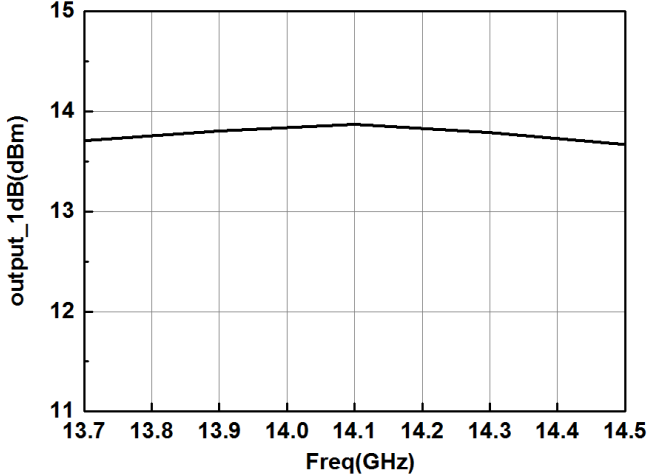


Figure 8. Output Power of the Single Channel (p-1dB)

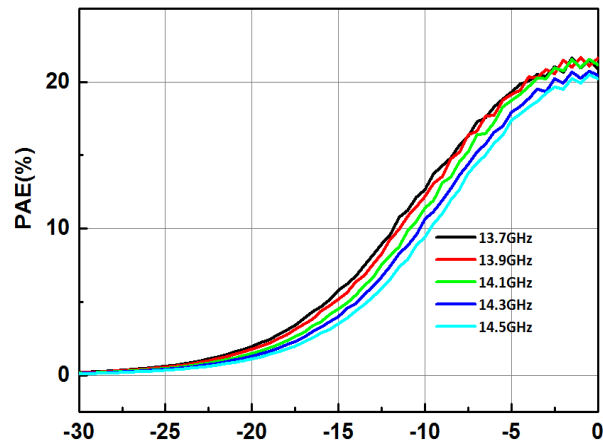


Figure 9. Relationship between Single-Channel Output Power and Efficiency

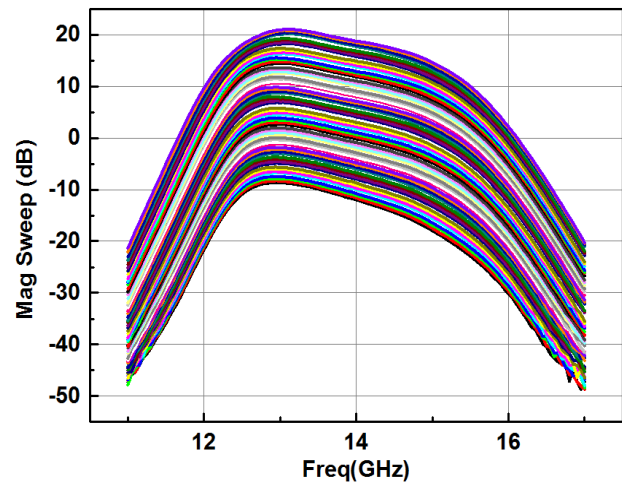


Figure 10. Amplitude Modulation (64-state)

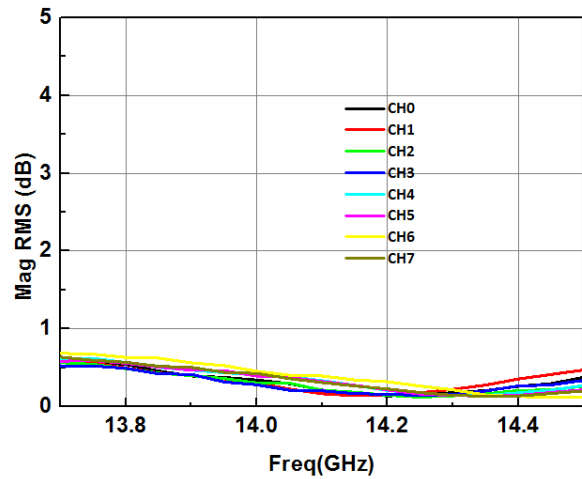


Figure 11. Relationship between 8 Channel Amplitude RMS and Frequency

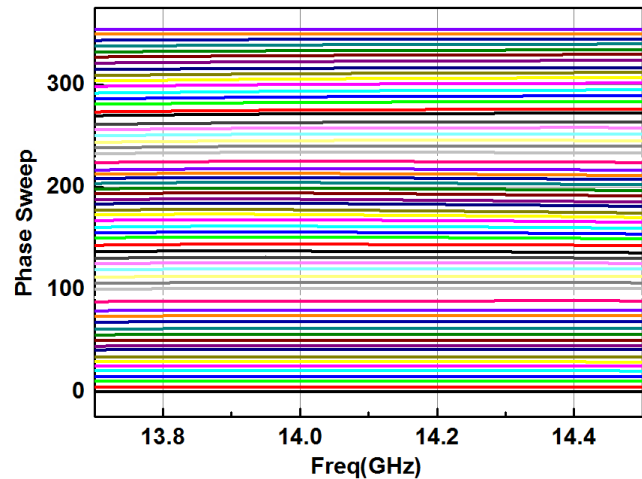


Figure 12. Relationship between 64 Phase Shift States and Frequency

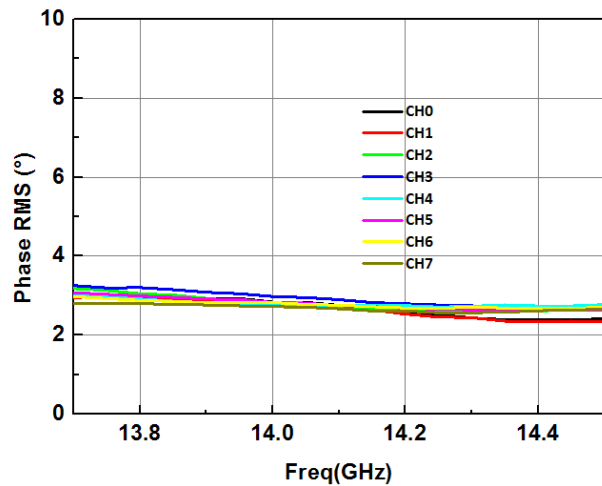


Figure 13. Relationship between 8 Channel Phase Shift RMS and Frequency

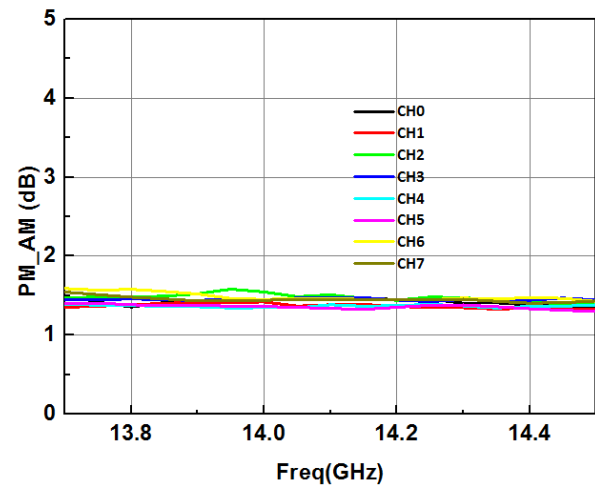


Figure 14. Parasitic Amplitude Modulation of Phase Shift

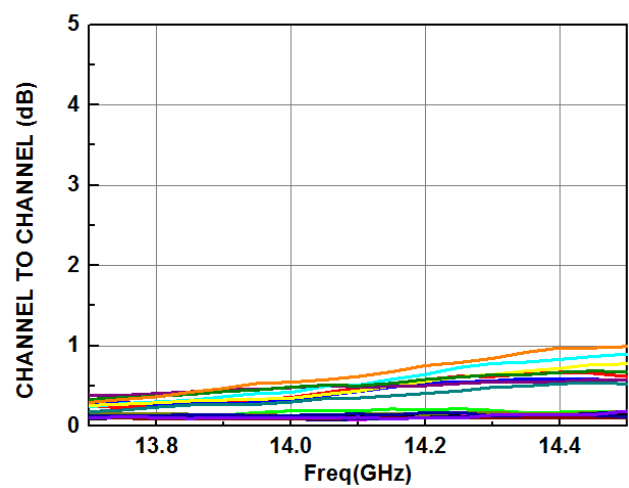


Figure 15. Amplitude Influence between Different Channels

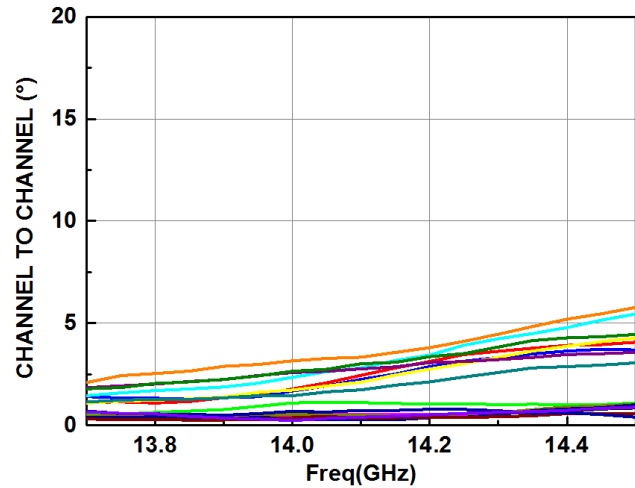


Figure 16. Phase Position Influence between Different Channels

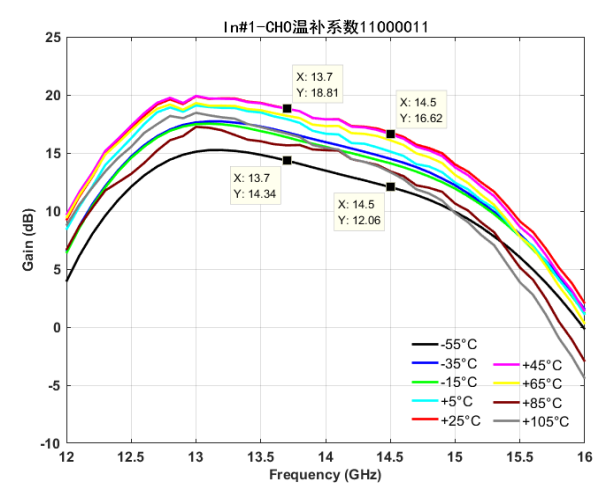


Figure 17. Gain Variations under High and Low Temperatures (-55°C - +105°C)

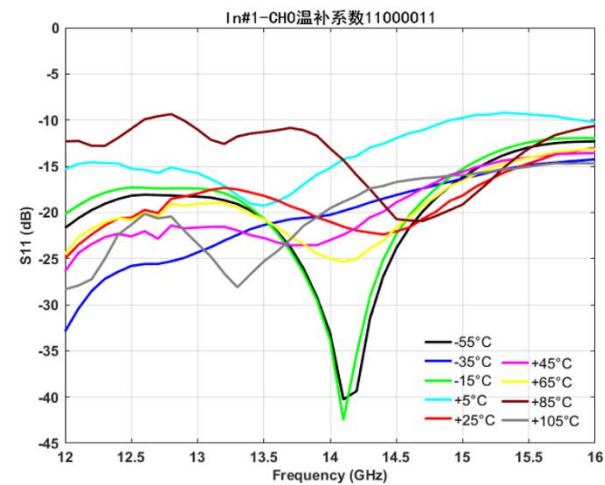


Figure 18. Input Echo Variations under High and Low Temperatures (-55°C - +105°C)

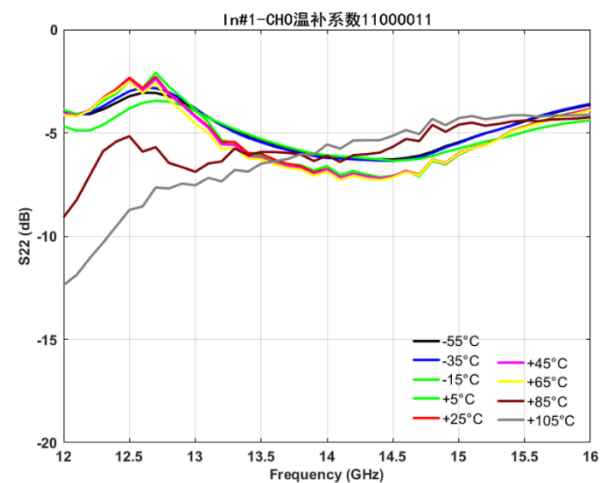


Figure 19. Output Echo Variations under High and Low Temperatures (-55°C - +105°C)

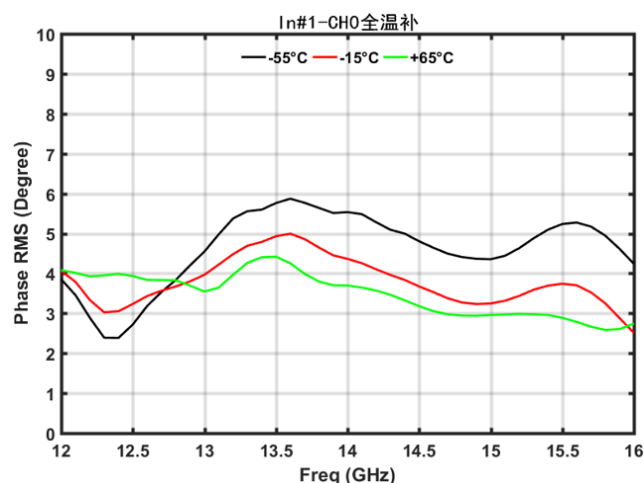


Figure 20. Phase Shift Accuracy Variations under High and Low Temperatures (-55°C - +105°C)

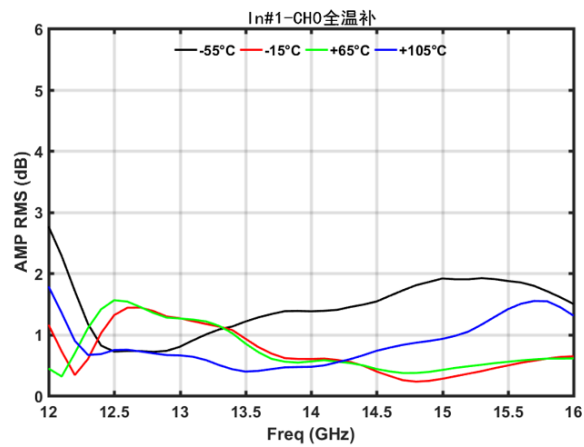


Figure 21. Amplitude Modulation Variations under High and Low Temperatures (-55°C - +105°C)

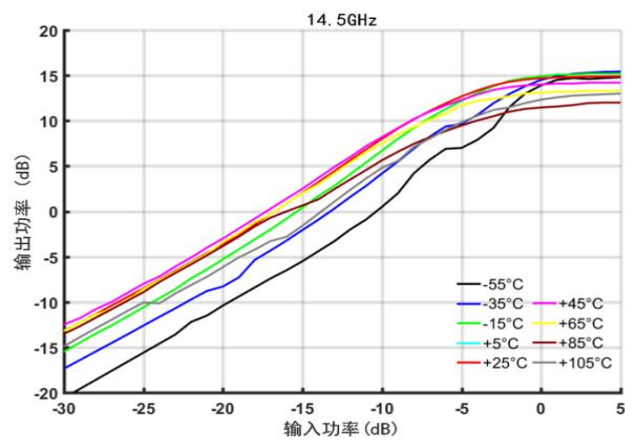


Figure22. Output Power Variations under High and Low Temperatures (-55°C - +105°C)

Chip Digital Control

Introduction to Digital Functions

The operating principle, communication protocol and application method of the SPI digital interface in the TRHJ-2041 chip are introduced in the following chapters.

Two kinds of power supplies and TTL(3.3V) levels are required for SPI digital modules, with 3.3V power supply for IO modules, 1.0V power supply for core modules and TTL(3.3V) level for IO signals in SPI digital modules.

The module has the following characteristics:

- Control the power supply status of 8 RF channels and support reading and writing.
- Read and write 8 RF channels with 6-bit amplitude and 6-bit phase.
- Read and write calibration codes for 64-level amplitude and 64-level phase.
- Supports reading and writing up to 64 8-bit on-chip control data (such as offset, etc.).
- Supports reading and writing up to 8 12-bit on-chip signal data (such as temperature, etc.).
- Adopt pipelined read-write mode, with large throughput.

Introduction to SPI interface

Table 4 Chip Interface Function

Interface Name	Function	Electrical level standard
LDAC_N	Load the amplitude and phase already written, and the falling edge is valid.	TTL (3.3V)
ID[3:0]	4-bit Chip ID	TTL (3.3V)
MODE	Switching between ID mode and ID-free mode.	TTL (3.3V)
SDO	Chip SPI output	TTL (3.3V)
RST_IN	Reset signal, low-level effective (256 clock synchronous reset).	TTL (3.3V)
SYNC	SPI chip selection, low-level effective	TTL (3.3V)
SDI	Chip SPI input	TTL (3.3V)
SCLK	Chip SPI clock	TTL (3.3V)
VDD_3.3	IO module power supply	3.3 V ($\pm 10\%$)
VDD_1.0	Core module power supply	1.0 V ($\pm 10\%$)

MODE: the switch of chip ID MODE. When Mode is 1 (pulled up to 3.3 V), SPI works in ID state. After ID is correctly configured, it can support reading and writing. When MODE is 0 (grounded), SPI works in an ID-free state, at which time the ID bit configuration is invalid, and this state only supports write operation. (See "chip ID configuration" for details of ID bit configuration.).

RST_IN: reset signal, low-level effective. When it remains low for at least 128 clock cycles, it is pulled high and

SPI is reset. To ensure the normal operation of the chip, reset operation should be performed first after the chip is powered on.

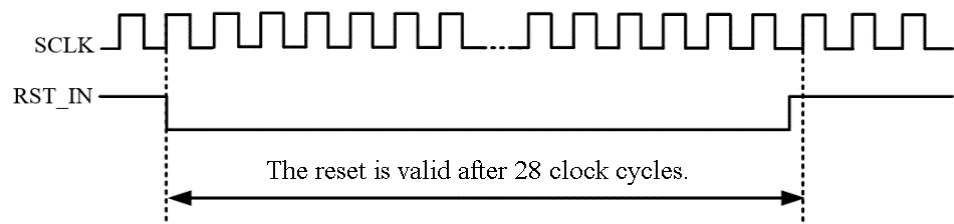


Figure 23. SPI Reset Timing

SPI Hardware Topology

The topological structure of the SPI hardware is as shown in Figure 24. The RF front-end chip is equipped with a 4-bit chip ID (ID[3:0]). One CS interface of the external MCU can control at most 16 RF front-end chips. Therefore, this RF front-end chip can realize the control of one master and multiple slaves. The number of slave control is determined by the number of master chip selection interfaces (16×the number of master chip selection interfaces).

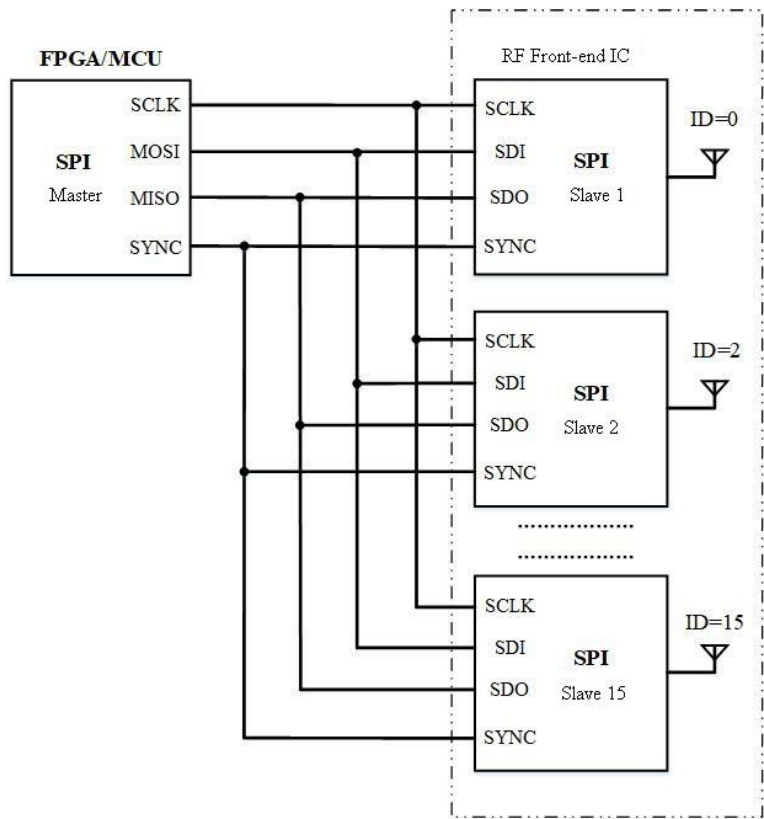


Figure 24. SPI Hardware Topology Diagram

Chip ID Configuration

The chip ID configuration only takes effect in the ID MODE (mode is 1), and the ID configuration methods of chips 0~ 15 are shown in the following table:

Table 5. Chip ID Configuration Table

D0~D3	Chip selection	D0~D3	Chip selection	D0~D3	Chip selection	D0~D3	Chip selection
-------	----------------	-------	----------------	-------	----------------	-------	----------------

0000	Chip 0	0100	Chip 4	1000	Chip 8	1100	Chip 12
0001	Chip 1	0101	Chip 5	1001	Chip 9	1101	Chip 13
0010	Chip 2	0110	Chip 6	1010	Chip 10	1110	Chip 14
0011	Chip 3	0111	Chip 7	1011	Chip 11	1111	Chip 15

Time Sequence

SPI communication protocol interface includes SCLK, SYNC, SDI and SDO. Read and write operations can be carried out. SCLK is the system synchronous clock interface of SPI master and slave. SYNC is an active low-level chip selection synchronization signal interface; SDI is the interface for host output (to slave); SDO is the interface of master input (from slave). The input and output are in frames, and the length of each frame is fixed at 24 bits.

Write Timing

As shown in Figure 25, when SYNC is lowed, the chip is selected. The serial input signal of SDI starts to be saved in the 24-bit buffer in the chip. Until all the 24-bit signals are stored, when SYNC changes from low to high, data processing is started, corresponding registers are read and written, and SDO output is prepared. When SYNC is high, the rising edge of SCLK latches the data into the corresponding register.

For the amplitude-phase control mode, after the data is latched into the corresponding registers on the rising edge of SCLK, the falling edge of LDAC_N loads the amplitude and phase data, making the amplitude and phase valid.

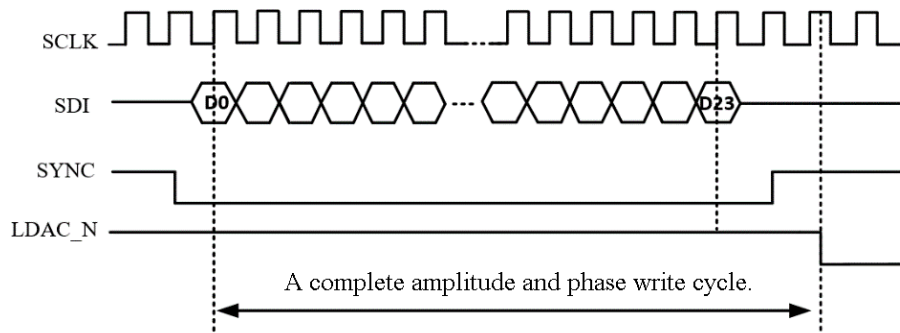


Figure 25. SPI Write Timing

Note:

- (1) To ensure the normal operation of the chip, after the current frame is sent and SYNC is pulled high, at least two SCLK rising edges should be separated before reading and writing the next frame;
- (2) To ensure the normal operation of the chip, it is required that there should be at least one SCLK period between the rising edge of SYNC and the falling edge of LDAC _ N.

Read Timing

The read timing only exists in ID MODE, that is, when mode is 1 (pulled up to 3.3V). The process is shown in

Figure 26. On the falling edge of SYNC, SDI writes a 24-bit read command frame. On the rising edge of SYNC, the output content is put into SDO buffer in advance. On the next falling edge of SYNC, SDO starts to output the corresponding read-back data. At the same time, the operation of writing the 24-bit read command frame by the next SDI will also take effect. If the next frame does not need to take effect, an invalid frame with the correct ID number can be written (for example, fill all with 1 after ID). Note that the ID number of the written frame must be correct, otherwise an error will occur in the output. At this time, the SDO state of read command frame is not concerned, because the current SDO state is only related to the previous frame.

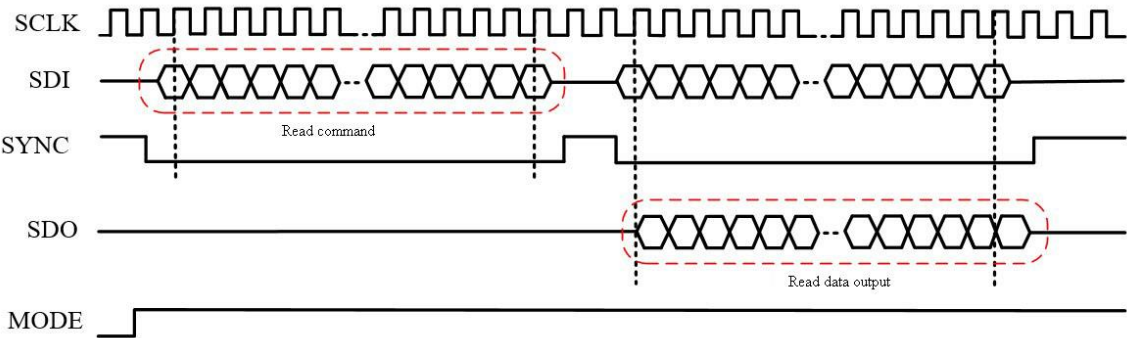


Figure 26. SPI Read Timing

Continuous reading and writing work reflect the characteristics of an assembly line. Taking Figure 27 as an example, the readout data of reading command 1 will be given when the reading command 2 is written in.

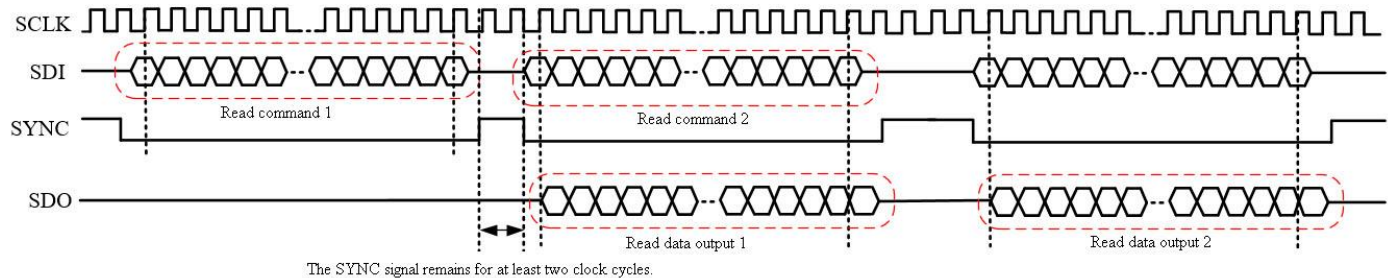


Figure 27. SPI Pipeline Schematic

Communication Protocol

Frame Composition

SPI communication is based on a fixed 24-bit frame. The overall frame structure is shown in Table 6. It can complete the functions of reading and writing channel power switch, amplitude and phase reading and writing, phase table reading and writing, on-chip control reading and writing, on-chip signal reading, pre-stored reading and writing, fast loading and pre-storing, etc.

The write command needs to be sent in a complete 24-bit frame. The read command only needs to send the frame

to the end of the address, that is, the bold black dividing line position in the table, and the following data part need not be sent, and can be filled with 0 or 1, and SDO will return a complete readback frame in the next frame.

SPI has a simple recognition mechanism for error frames. When the structure of the input frame is larger than 24 bits, the frame is judged as an error frame and is directly discarded.

Table 6. Overall Frame Structure

Sending order	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
Channel switch	Chip ID			Read and write	Function selection			NULL							Channel number									
	MSB		LSB	Read 1 and write 0	0	0	0								MSB LSB									
Amplitude e phase	Chip ID			Read and write	Function selection			NULL	Channel selection			Amplitude				Phase								
	MSB		LSB	Read 1 and write 0	0	0	1		MSB		LSB	MSB LSB			MSB LSB									
Phase meter	Chip ID			Read and write	Function selection			Phase table address					MLSB		NULL	Data								
	MSB		LSB	Read 1 and write 0	0	1	1	MSB LSB							MSB LSB									
Signal Register	Chip ID			Read and write	Function selection				Register address			Register data												
	MSB		LSB	Read only 1	0	1	0	1	MSB		LSB	MSB LSB												
Control Register	Chip ID			Read and write	Function selection					Register address					Register data									
	MSB		LSB	Read 1 and write 0	0	1	0	0	1	MSB LSB					MSB LSB									

Register Function and Address Mapping.

Channel Supply Switch Register

Channel switch register can control the power supply switches of 8 RF channels. Eight bits correspond to eight channels, the highest bit corresponds to channel 7, and the lowest bit corresponds to channel 0. High level on, low level off.

Table 7. Channel Switch Register Frame Structure

Chip ID				Read and write	Function			NULL								Channel number									
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23		
MSB		LSB		Read 1 and	0	0	0	NULL								MSB								LSB	

(1) Read/write configuration: when performing read/write configuration, D4 is 1 for reading operation and D4 is 0 for writing operation.

(2) Function selection configuration: D5~D7 are function selection bits, and D5~D7 must be [000] when configuring the channel power supply switch register.

(3) Idle bit configuration: D8~D15 are idle bits which can be filled with 0 or 1.

(4) Channel number configuration: D16~D23 are channel number control bits, and the truth value table is as follows.

Table 8. Channel Number Configuration Truth Value Table

D16	D17	D18	D19	D20	D21	D22	D23	Channel Status
0	0	0	0	0	0	0	0	Channel fully closed
0	0	0	0	0	0	0	1	Channel 0 open
0	0	0	0	0	0	1	0	Channel 1 open
0	0	0	0	0	1	0	0	Channel 2 open
0	0	0	0	1	0	0	0	Channel 3 open
0	0	0	1	0	0	0	0	Channel 4 open
0	0	1	0	0	0	0	0	Channel 5 open
0	1	0	0	0	0	0	0	Channel 6 open
1	0	0	0	0	0	0	0	Channel 7 open
1	1	1	1	1	1	1	1	Channel fully open

Example: in ID MODE (mode is 1), carry out power supply write operation to channel 1 of chip 1(ID is 1).

The corresponding 24-bit frame should be configured as: D0~D23[0001_0_000_11111111_00000010].

Amplitude Phase Register

The amplitude register can configure the amplitude and phase of the designated channel. In which the amplitude and phase configurations each occupy 6 bits.

Table 9. Frame Structure of Amplitude and Phase Register

Chip ID				Read and	Function selection			NULL	Channel selection			Amplitude						Phase					
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
MSBLSB				Read 1	0	0	1	NULL	MSBLSB			MSBLSB						MSBLSB					

(1) Channel selection configuration: D9~D11 are channel selection control bits, and the truth table is as follows.

Table 10. Channel Selection Truth Table of Amplitude and Phase Register.

Channel number	Channel 0	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
D9	0	0	0	0	1	1	1	1
D10	0	0	1	1	0	0	1	1
D11	0	1	0	1	0	1	0	1

(2) Amplitude configuration: Amplitude control occupies 6 bits in total, and D12 (the highest bit) - D17 (the lowest bit) corresponds to 16dB - 0.5 dB respectively.

Table 11. Amplitude Configuration Truth Table.

D12~D17	Amplitude modulation value (dB)	D12~D17	Amplitude modulation value (dB)	D12~D17	Amplitude modulation value (dB)	D12~D17	Amplitude modulation value (dB)
00_0000	0	01_0000	8	10_0000	16	11_0000	24
00_0001	0.5	01_0001	8.5	10_0001	16.5	11_0001	24.5
00_0010	1	01_0010	9	10_0010	17	11_0010	25
00_0011	1.5	01_0011	9.5	10_0011	17.5	11_0011	25.5
00_0100	2	01_0100	10	10_0100	18	11_0100	26
00_0101	2.5	01_0101	10.5	10_0101	18.5	11_0101	26.5
00_0110	3	01_0110	11	10_0110	19	11_0110	27
00_0111	3.5	01_0111	11.5	10_0111	19.5	11_0111	27.5
00_1000	4	01_1000	12	10_1000	20	11_1000	28
00_1001	4.5	01_1001	12.5	10_1001	20.5	11_1001	28.5
00_1010	5	01_1010	13	10_1010	21	11_1010	29
00_1011	5.5	01_1011	13.5	10_1011	21.5	11_1011	29.5
00_1100	6	01_1100	14	10_1100	22	11_1100	30
00_1101	6.5	01_1101	14.5	10_1101	22.5	11_1101	30.5
00_1110	7	01_1110	15	10_1110	23	11_1110	31
00_1111	7.5	01_1111	15.5	10_1111	23.5	11_1111	31.5

(3) Phase configuration: Phase control occupies 6 bits, and D18 (the highest bit) - D23 (the lowest bit) corresponds to 180 ° - 5.625 respectively.

Table 12. Phase Configuration Truth Table

D18~D23	Phase Modulation Value (°)	D18~D23	Phase Modulation Value (°)	D18~D23	Phase Modulation Value (°)	D18~D23	Phase Modulation Value (°)
00_0000	0	01_0000	90	10_0000	180	11_0000	270
00_0001	5.625	01_0001	95.625	10_0001	185.625	11_0001	275.625
00_0010	11.25	01_0010	101.25	10_0010	191.25	11_0010	281.25
00_0011	16.875	01_0011	106.875	10_0011	196.875	11_0011	286.875
00_0100	22.5	01_0100	112.5	10_0100	202.5	11_0100	292.5
00_0101	28.125	01_0101	118.125	10_0101	208.125	11_0101	298.125
00_0110	33.75	01_0110	123.75	10_0110	213.75	11_0110	303.75
00_0111	39.375	01_0111	129.375	10_0111	219.375	11_0111	309.375
00_1000	45	01_1000	135	10_1000	225	11_1000	315
00_1001	50.625	01_1001	140.625	10_1001	230.625	11_1001	320.625

00_1010	56.25	01_1010	146.25	10_1010	236.25	11_1010	326.25
00_1011	61.875	01_1011	151.875	10_1011	241.875	11_1011	331.875
00_1100	67.5	01_1100	157.5	10_1100	247.5	11_1100	337.5
00_1101	73.125	01_1101	163.125	10_1101	253.125	11_1101	343.125
00_1110	78.75	01_1110	168.75	10_1110	258.75	11_1110	348.75
00_1111	84.375	01_1111	174.375	10_1111	264.375	11_1111	354.375

Control Register

There are 64 8-bit control registers in this chip. This category of register channel information is contained in the register address. The address and function mapping of the control register are shown in Table 14.

Table 13. Control Register Frame Structure

Chip ID				Read and write	Function selection					Register address						Register data							
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23
MSB		LSB		Read 1 and	0	1	0	0	1	MSB						LSB							

Table 14. Control Register Address Mapping

No.	Function	Register address	Register initial value	Description
Chip bias				
1	CH0_PD bias	CTRL00[5:0]	0010_0000	
2	CH0_PA bias	CTRL01[5:0]	0010_0000	
3	CH0_AMP2 bias	CTRL02[5:0]	0010_0000	
4	CH0_AMP1 bias	CTRL03[5:0]	0010_0000	
5	CH2_PD bias	CTRL04[5:0]	0010_0000	
6	CH2_PA bias	CTRL05[5:0]	0010_0000	
7	CH2_AMP2 bias	CTRL06[5:0]	0010_0000	
8	CH2_AMP1 bias	CTRL07[5:0]	0010_0000	
9	CH4_PD bias	CTRL08[5:0]	0010_0000	
10	CH4_PA bias	CTRL09[5:0]	0010_0000	
11	CH4_AMP2 bias	CTRL10[5:0]	0010_0000	
12	CH4_AMP1 bias	CTRL11[5:0]	0010_0000	
13	CH6_PD bias	CTRL12[5:0]	0010_0000	
14	CH6_PA bias	CTRL13[5:0]	0010_0000	
15	CH6_AMP2 bias	CTRL14[5:0]	0010_0000	
16	CH6_AMP1 bias	CTRL15[5:0]	0010_0000	
17	CH1_PD bias	CTRL17[5:0]	0010_0000	
18	CH1_PA bias	CTRL18[5:0]	0010_0000	
19	CH1_AMP2 bias	CTRL19[5:0]	0010_0000	
20	CH1_AMP1 bias	CTRL20[5:0]	0010_0000	
21	CH3_PD bias	CTRL21[5:0]	0010_0000	
22	CH3_PA bias	CTRL22[5:0]	0010_0000	
23	CH3_AMP2 bias	CTRL23[5:0]	0010_0000	
24	CH3_AMP1 bias	CTRL24[5:0]	0010_0000	
25	CH5_PD bias	CTRL25[5:0]	0010_0000	

No.	Function	Register address	Register initial value	Description
26	CH5_PA bias	CTRL26[5:0]	0010_0000	
27	CH5_AMP2 bias	CTRL27[5:0]	0010_0000	
28	CH5_AMP1 bias	CTRL28[5:0]	0010_0000	
29	CH7_PD bias	CTRL29[5:0]	0010_0000	
30	CH7_PA bias	CTRL30[5:0]	0010_0000	
31	CH7_AMP2 bias	CTRL31[5:0]	0010_0000	
32	CH7_AMP1 bias	CTRL32[5:0]	0010_0000	

Note: the address of the control register is controlled by D10-D15, and the decimal should be converted into binary in SPI read-write communication frame. For example, CTRL10[5:0] is written as 00_1010.

Example: Under the mode with ID (mode of 1), PA bias of the Channel 0 of Chip 1 (ID of 1) should be provided with writing operation.

The corresponding 24-bit frame should be configured as: D0-D23[0001_0_01001_000001_00100000].

Temperature Coefficient Control Register

Table 15. Temperature Coefficient Control Register Address Mapping

No.	Function	Register address	Register initial value	Description
Temperature Compensation Coefficient				
1	AMP1 temperature coefficient	CTRL51[7:0]	1111_0000	1111_0000 corresponds to the maximum temperature compensation state, and 0000_1111 corresponds to the minimum temperature compensation state.
2	Temperature coefficient of power detection	CTRL52[7:0]	1111_0000	
3	Temperature coefficient of power amplifier	CTRL53[7:0]	1111_0000	
4	AMP2 temperature coefficient	CTRL54[7:0]	1111_0000	

Temperature compensation is controlled by registers P and Z respectively. the first four bits in the initial value of the register are the value of P, the last four bits are the value of Z, and the value of register Z is the bitwise inversion of P. therefore, there are 16 temperature compensation states.

It is recommended that the temperature coefficient be configured in full temperature compensation state (that is, the values of the temperature coefficient control registers are all set according to the initial values in the table). Refer to Table 13 for 24-bit frame structure.

Phase Table Register

SPI prestores a default phase control table, and the phase shift performance can be optimized by writing the optimized phase control table. The phase control table is a 64×24-bit table (TXT file, see the phase table file for details). Every 24 bits (one line) represents a phase state, with a total of 64 phase states. Each phase state is divided into four writes, with 6 bits of data written each time.

Table 16. Phase Table Register Frame Structure

Chip ID				Read and write		Function selection			Phase table address						MLSB		NULL		Data									
D0	D1	D2	D3	D4		D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23				
MSB				LSB		Read 1 and write 0		0	1	0	MSB						LSB		NULL		MSB						LSB	

(1) Address configuration: the frame has 6 bits of addresses D8-D13, which are 0-63 in binary, and there are 64 phase state addresses in total.

(2) MLSB configuration: This function is controlled by D14 - D15. Each phase state is 24 bits, which is written in four times. 00 refers to the last 6 bits, 01 refers to the middle-back 6 bits, 10 refers to the middle-front 6 bits, and 11 refers to the front 6 bits.

(3) Idle bit configuration: D16-D17 are idle bits which can be filled with 0 or 1.

(4) Data configuration: D18-D23 refer to the phase table data configuration, see the phase table file for details.

Signal Register

The signal register is in read-only mode, and only SIG07 is a valid signal register. The address and function mapping are shown in Table 18, and 10 bits (D12 and D13 are invalid bits) of ADC output temperature and power data can be read out.

Table 17. Signal Register Frame Structure

Chip ID				Read and	Function selection				Register address			Register data															
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23				
MSB				LSB	Read only 1	0	1	0	1	MSB		LSB	MSB										LSB				

Table 18. Signal Register Address Mapping

Address	Name	Function
SIG07[9]	Q9	ADC outputs MSB bits in parallel
SIG07[8]	Q8	ADC outputs in parallel
SIG07[7]	Q7	ADC outputs in parallel
SIG07[6]	Q6	ADC outputs in parallel
SIG07[5]	Q5	ADC outputs in parallel
SIG07[4]	Q4	ADC outputs in parallel
SIG07[3]	Q3	ADC outputs in parallel
SIG07[2]	Q2	ADC outputs in parallel
SIG07[1]	Q1	ADC outputs in parallel
SIG07[0]	Q0	ADC outputs LSB bits in parallel

Note: Before reading the 10-bit ADC output temperature and power data in the signal register, the user needs to write the 24-bit frame in the control registers CTRL62 and CTRL63. Then carry out the read-only operation to the signal register SIG07. (see "Temperature Readback Operation" and "Power Readback Operation" for details.)

Temperature and Power Readback

ADC Control Register

Table 19. Temperature Sensor and ADC Control Register Address Mapping

No.	Function	Register address	Register initial value	Description
ADC & power detection & temperature sensor				
1	ADC input selection	CTRL62[4:0]	0_0000	Initial value 0_0000 of register: Input temperature; 0_0000-0_1001: Input CH0-CH7 power detection. RX does not have such function.

No.	Function	Register address	Register initial value	Description
4	Temperature verification & calibration RES[5:0]	CTRL63[5:0]	10_0000	1 bit corresponds to a temperature change of 0.4 degrees.

Temperature Readback Operation

Step 1: Perform RST_IN reset operation after the chip is powered on;

Step 2: As shown in Table 19, write the 24-bit frames of CTRL62 and CTRL63 control registers in turn;

Step 3: Wait for at least 2400 SCLK clock cycles;

Step 4: Write a 24-bit frame for reading back the SIG07 signal register;

Step 5: Read back the return value of the SIG07 signal register to obtain the temperature.

$$\text{Temperature calculation: Temp.} = 249 - 0.364 * \text{Code}_{(D)}$$

Example: when the temperature verification calibration RES[5:0] = 10_0000 (default), the ADC output corresponding to 27°C is 10_0110_0100 (610); When the temperature calibration RES is 11_1111, the ADC output corresponding to 27°C is 10_1000_0100(644).

Table 20. Temperature vs ADC Output

Temperature [°C]	-40	-20	0	20	27	40	60	80	100
ADC output (D)	794	739	684	629	610	574	519	464	408

There is an error (maximum 12°C) in the corresponding temperature of the same ADC reading, and the relative difference is unchanged. It can be compensated by temperature verification&calibration RES[5:0].

Power Readback Operation

Step 1: Perform RST_IN reset operation after the chip is powered on;

Step 2: Configure the chip according to the chip write operation to ensure the normal operation of the chip;

Step 3: As shown in Table 19, write the 24-bit frames of CTRL62 and CTRL63 control registers in turn;

Step 4: Wait for at least 2400 SCLK clock cycles;

Step 5: Write a 24-bit frame for reading back the SIG07 signal register;

Step 6: read back the return value of SIG07 signal register to get the power.

Output of ADC corresponding to power detection voltage: $\text{ADC}(\text{decimal}) = 1023 - 338V_{PD}$

Table 21. Correspondence between Power Detection Voltage and ADC Output

Pout [dBm]	0.00	1.41	2.63	3.86	5.07	6.28	7.45
Vout [mV]	45	53	74	119	218	363	522
ADC [H]	1009	1005	998	983	949	900	847
Pout [dBm]	8.59	9.67	10.66	11.54	12.27	12.87	13.36
Vout [mV]	664	746	782	801	814	822	829
ADC [H]	799	771	759	752	748	745	743

Chip SPI Configuration Steps

■ Chip Write Operation

Step 1: Perform RST_IN reset operation after the chip is powered on;

Step 2: Write each 24-bit frame corresponding to the phase table in turn (if the initial phase table does not meet the use requirements, this step can be performed; otherwise, this step can be skipped).

Step 3: Write 24-bit frames corresponding to different control registers in turn, and configure chip offset, temperature compensation, high-low speed switching, etc.

Step 4: Write a 24-bit frame corresponding to the channel switch register;

Step 5: Write a 24-bit frame corresponding to the T/R amplitude and phase control register;

■ Chip Read Operation

Step 1: Perform RST_IN reset operation after the chip is powered on;

Step 2: Write the 24-bit frame corresponding to each register for readback. (Please refer to "Temperature Readback Operation" and "Power Readback Operation" for ADC temperature and power readback operation).

Step 3: Read back the return value of each register.

Recommended RF Application and Design Process

■ Process recommendation and precautions

1. The matching plate in the figure below is RO4350B with a thickness of 0.168mm;
2. RF port impedance is 50 ohms;
3. RF circuit form (CPW) can be implemented according to the recommended size on the left;
4. Ensure that the packaged chip GSG has a good RF connection;
5. Pad design: The size of the chip solder ball is 250um; The recommended pad window size is 230um;
6. The products are welded by SMT, and the solder ball is a 217-degree SnAg ball. The maximum reflow soldering temperature shall not exceed 260 degrees;
7. Heat dissipation channels are added on the back of the chip to improve the heat dissipation performance of the device package to the greatest extent;
8. All pads of the chip need to be grounded on PCB;
9. When the chip is not in use, it is recommended to store it under the protection of vacuum or nitrogen;
10. Pre-baking is required before chip installation on PCB, and the baking temperature should not exceed 120°C;
11. The chip is sensitive to static electricity, so the whole process needs to be protected against static electricity. It is suggested to add anti-static design to PCB boards.

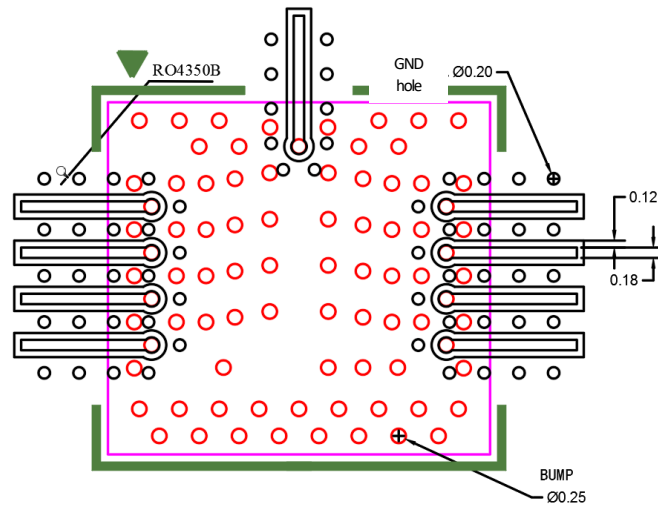


Figure 28. Recommended Application of RF SMT

Note:

(1) The port impedance of the TRHJ-2041 multi-function integrated chip is 50 ohms. Ensure that the RF GND of the chip can achieve good grounding continuity.

(2) It is needed to weld the naked bonding pad below the TRHJ-2041 to corresponding welding spots of the impedance to low heating temperature and the electrical impedance evaluation board. The ground pin should be connected to all other ground layers on the evaluation board to maximize the thermal performance of the device package.

Suggested Application Form

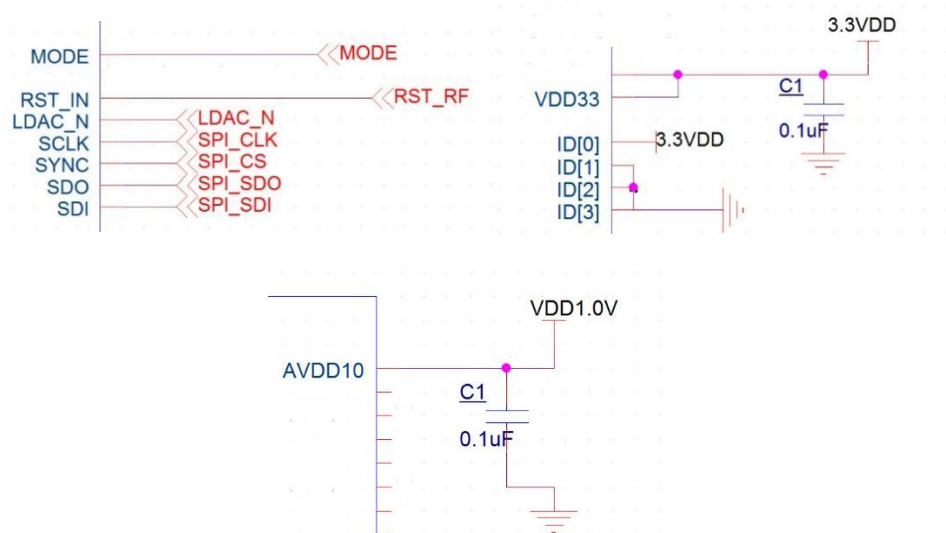


Figure 29. Recommended Peripheral Bias of Multifunctional Integrated Chip

Schematic Diagram and Layout Outline of Evaluation Board

A. Schematic of Evaluation Board

The evaluation board can support multi-channel testing. The circuit schematic diagram of the board is shown below.

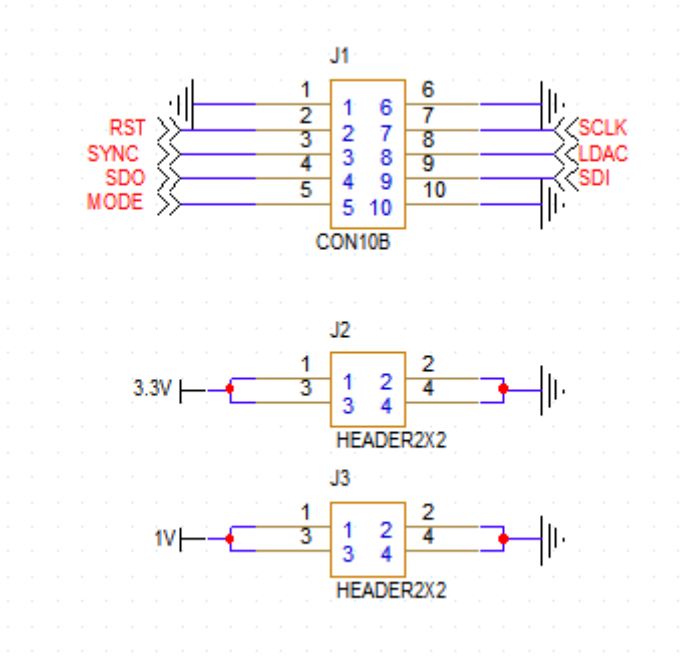


Figure 30. Schematic Diagram of Multifunctional Integrated Chip Evaluation Board

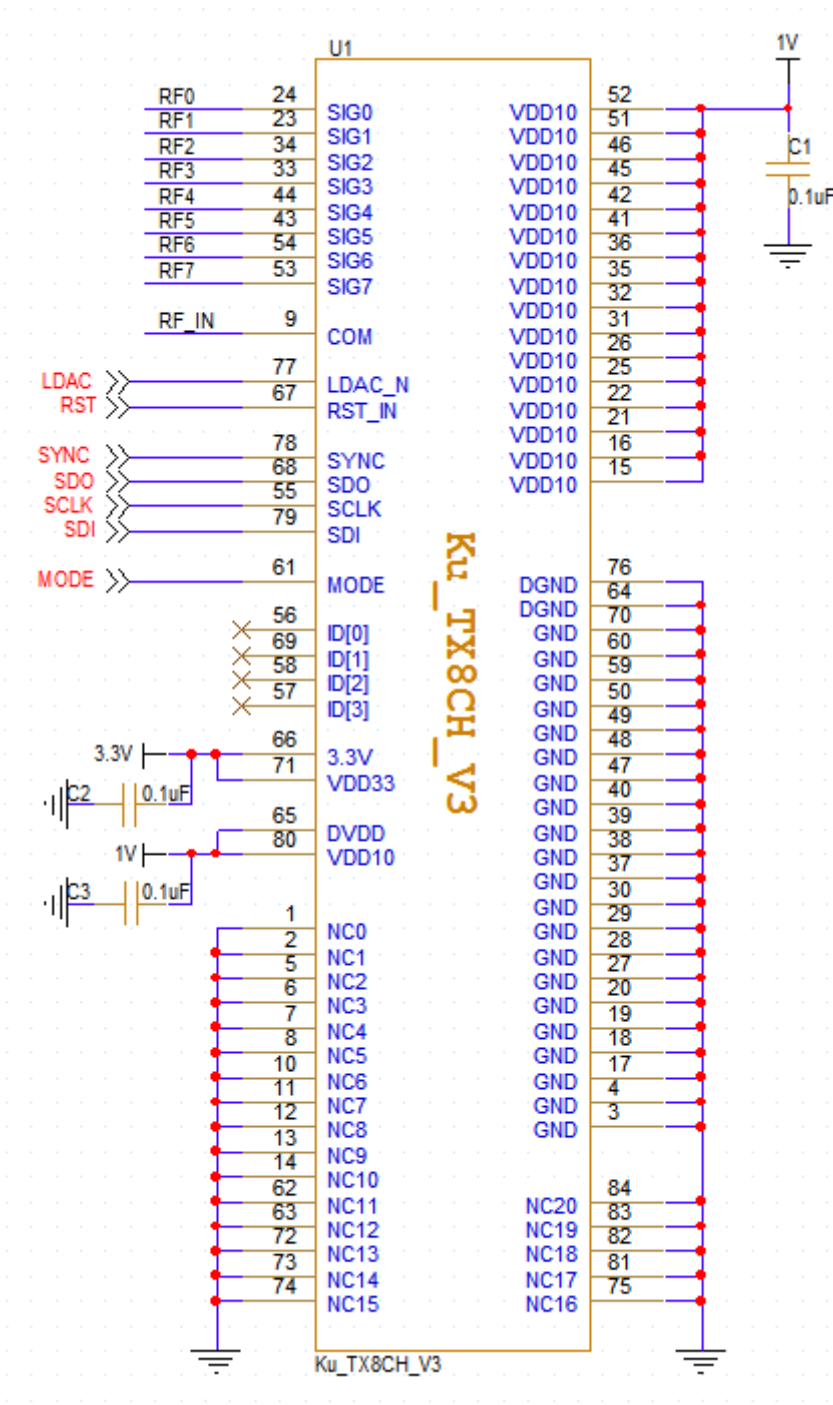


Figure 31. Schematic Diagram of Multifunctional Integrated Chip Evaluation Board

B. Circuit Layout of Evaluation Board

The evaluation board adopt Rogers4350B PCB in a four-layer board structure. The overlay information is as follows:

No.	Lamination Layer	Description	Thickness (mm)	Material
1	TOP	RF	0.035	copper
	/	/	0.1	Ro4350B
2	GND2	GND	0.035	copper

	/	/	0.168	Ro4350B
3	ART3	SPI / Power	0.035	copper
	/	/	0.168	Ro4350B
4	BOTTOM	GND	0.035	copper

TOP layer:

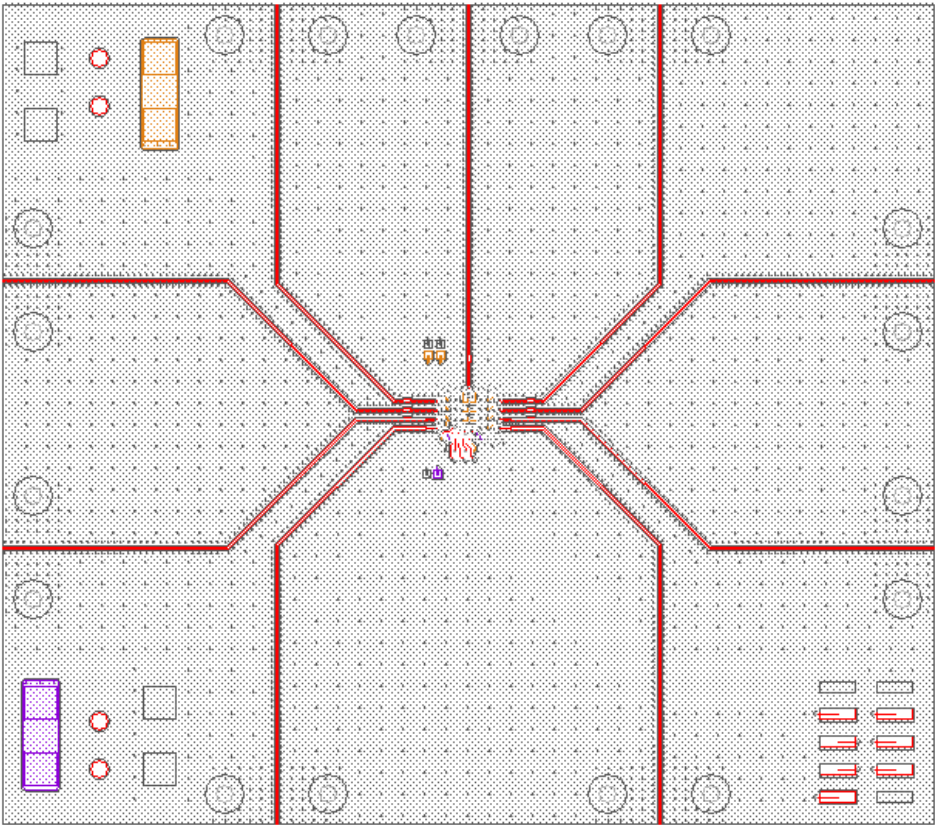


Fig. 32. Top Layer of Multifunctional Integrated Chip Evaluation Board

GND2 layer:

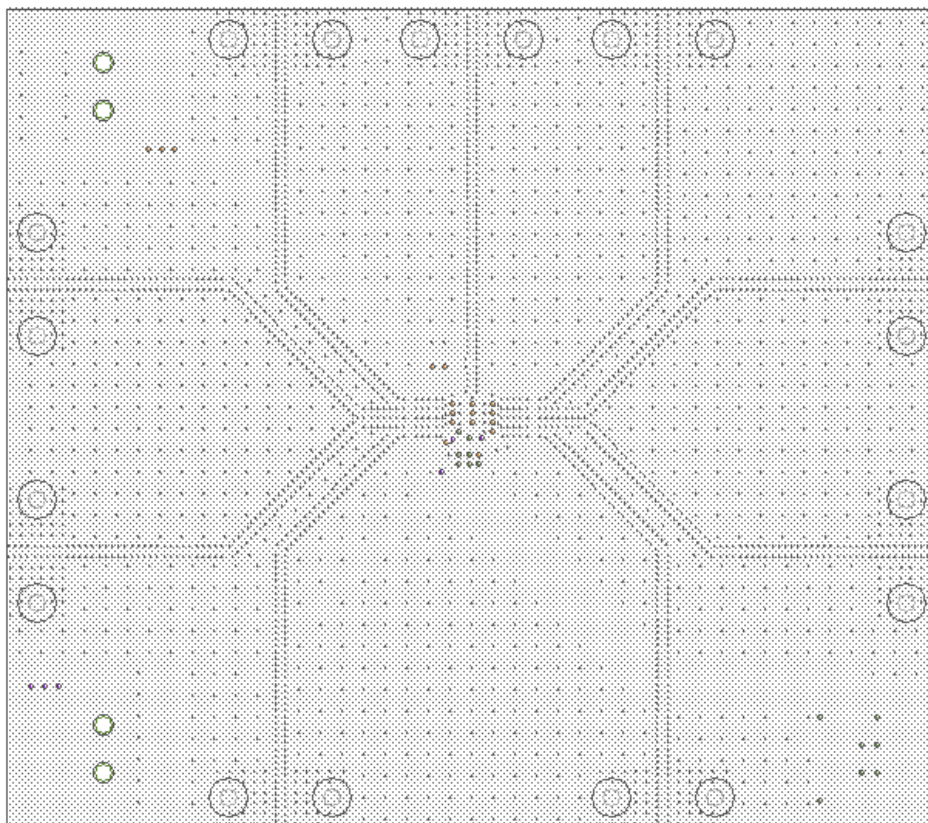


Fig. 33. L2 Layer of Multifunctional Integrated Chip Evaluation Board

ART3 layer:

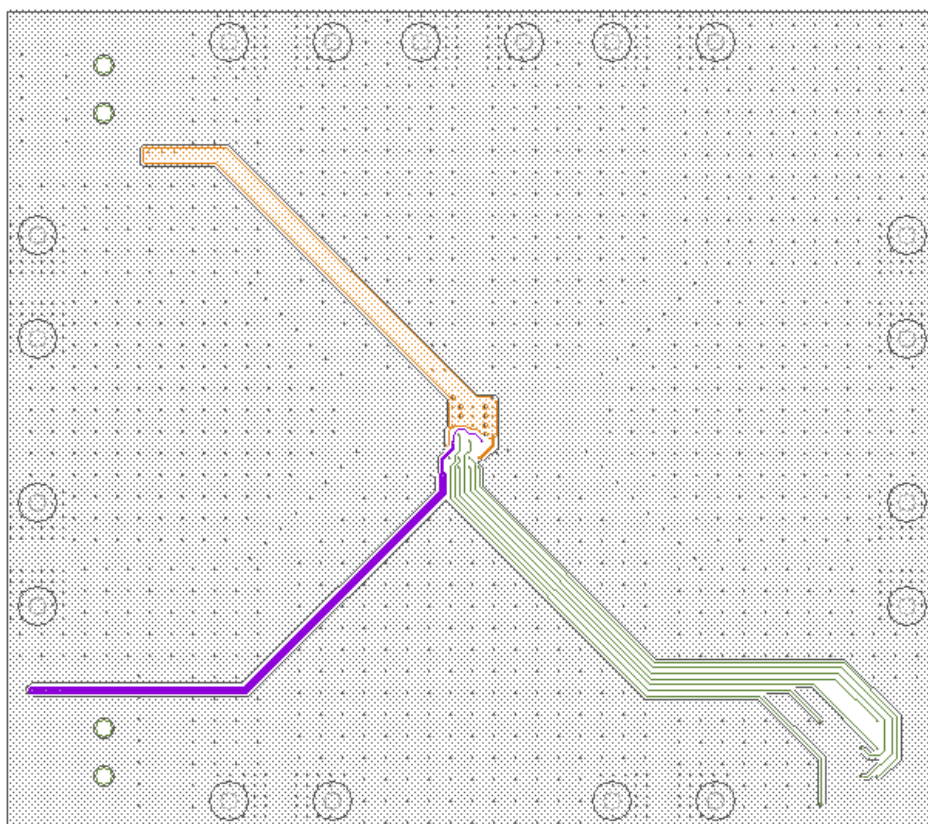


Figure 34. L3 Layer of Multifunctional Integrated Chip Evaluation Board

BOTTOM layer:

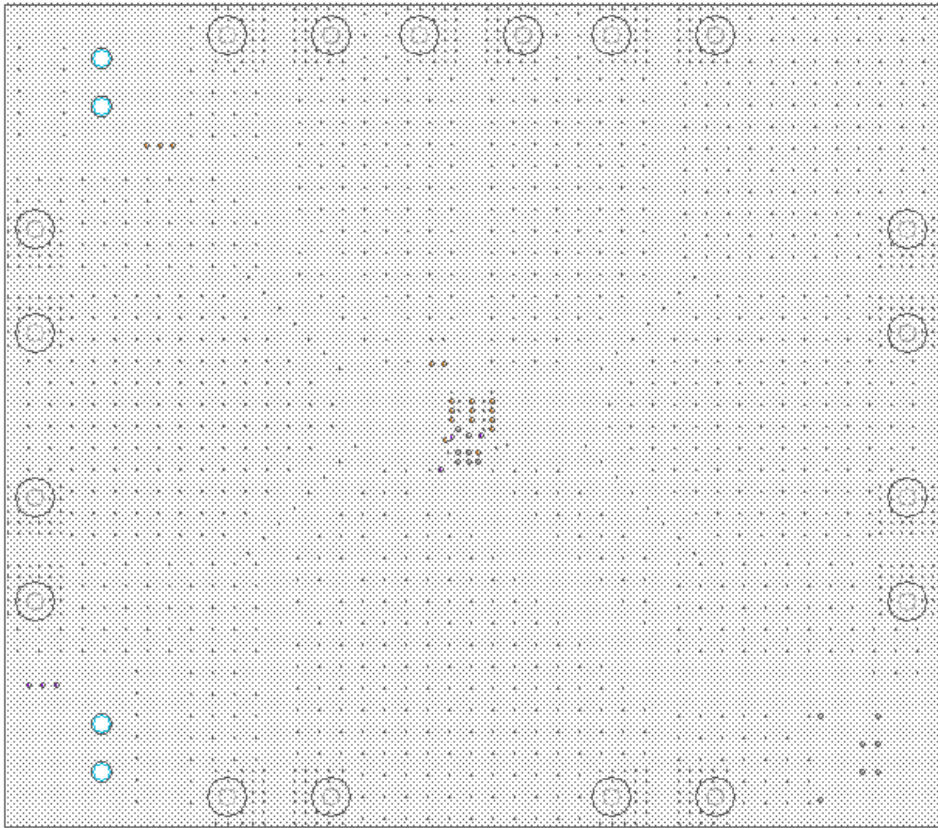


Figure 35. BOTTOM Layer of Multifunctional Integrated Chip Evaluation Board

F . Introduction to Evaluation Board Software

The chip evaluation software is a special control program developed by our company for chip evaluation board.

The functions of the software are mainly divided into three parts, including chip initialization, channel amplitude and phase control and readback. Through the evaluation software, with the evaluation board and control board, the basic technical indicators of the chip can be quickly tested and verified. In addition, the provided code examples in combination with SPI protocol allow users to develop related functions independently.



Figure 36. Evaluation Software Interface

For more evaluation board details and software applications, please refer to the development kit application notes.

Ordering Guide

Model	Temperature Range	Package Description	Delivery
TRHJ-2041-WLP	-50°C - +70°C	84PINWLCSP	
TRHJ-2041-BGA		84PINFC-BGA	
TRHJ-2041-LGA		44PINFC-LGA	